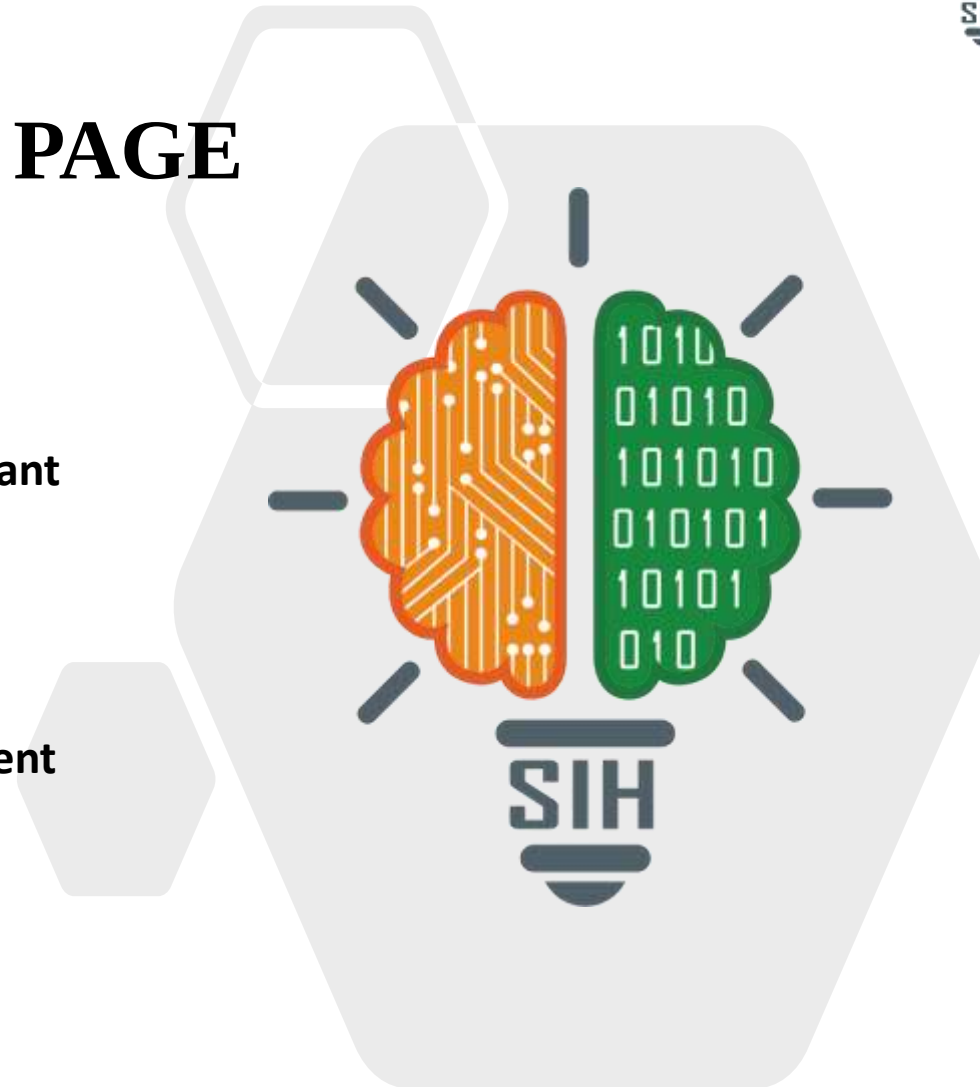


TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title - AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY
- Theme - Agriculture, Food Tech & Rural Development
- PS Category - Software
- Team ID -
- Team Name - Silicon Alley



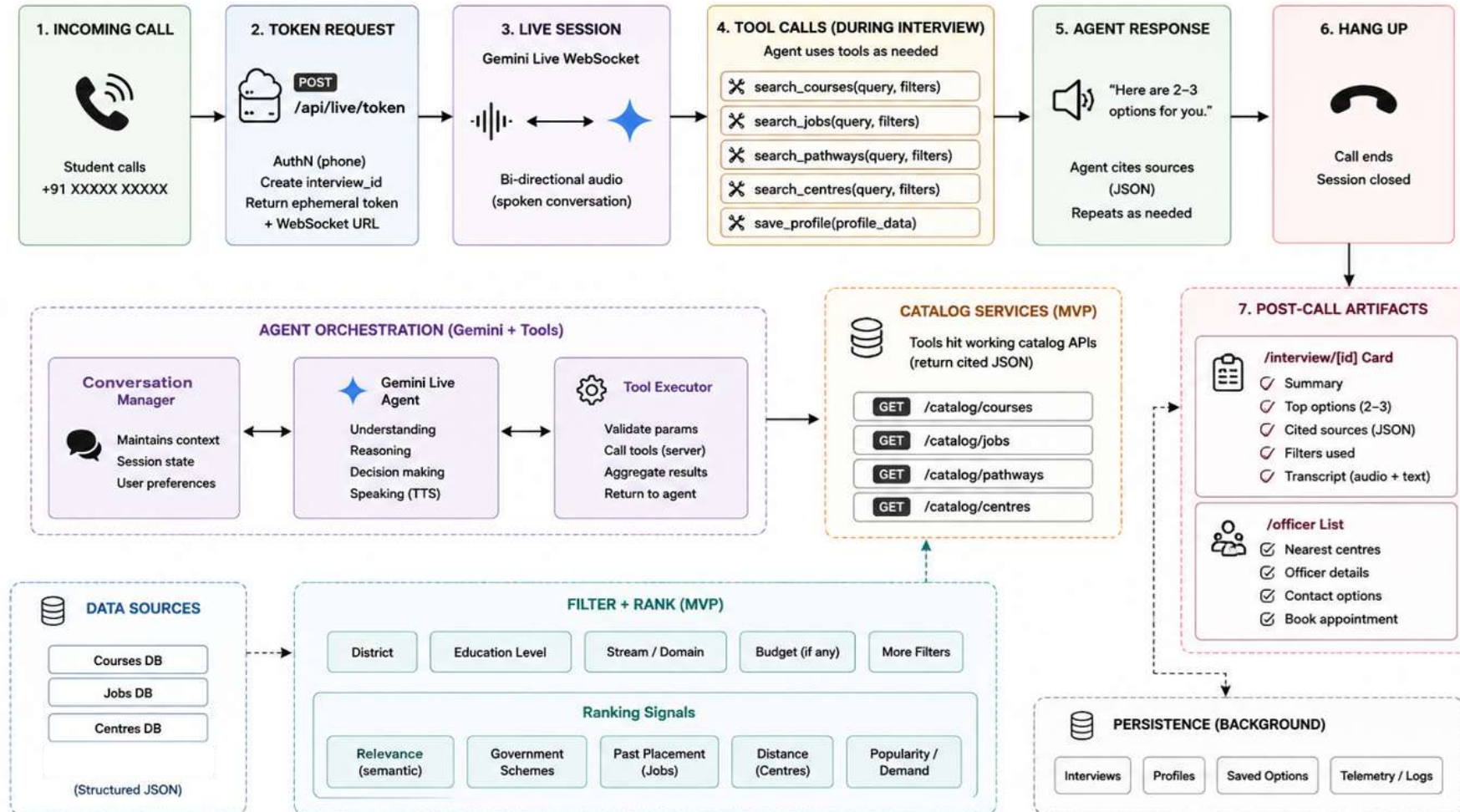
PM-AJAY Voice Livelihood Helpline

- **Proposed Solution:** Working phone-call prototype for PM-AJAY GIA counselling.
- **Detailed explanation of the proposed solution**
 - Beneficiary taps Call and speaks.
 - Gemini Live interviews in their language/dialect.
 - Collects education, family occupation, current work, skills, constraints, wage vs self-employment, district etc.
 - Recommends 2-3 nearby NSQF courses, jobs, or GIA/NSFDC pathways and saves a counselling record.
- **How it addresses the problem**

Form-based GIA intake fails low-literacy SC users; mismatched trades, dropouts, poor placement.
Voice interview + district catalog matches aspirations to local NSQF training and livelihood demand.
- **Innovation and uniqueness of the solution**
 1. Native voice-to-voice;
 2. no language picker;
 3. replies in the same dialect, respectfully.
 4. Agent tools search NQR, NCS, NSFDC, SIDH, PM-AJAY catalogs during the call.
 5. Officer desk turns each call into a structured GIA counselling file.

Technologies used

- Next.js 16, React 19, TypeScript, Tailwind 4, bun.
- Google Gemini Live (gemini-3.1-flash-live-preview) via @google/genai.
- Browser mic 16 kHz PCM; speaker 24 kHz.
- API key stays on server (ephemeral token).
- Hardware now: smartphone / kiosk browser.
- **Future: IVR feature phone, WhatsApp voice notes.**



FEASIBILITY AND VIABILITY

- **Analysis of the feasibility of the idea**
 - Multilingual live voice is Gemini; we did not train a model.
 - Catalog is official public lists (NQR/NCS/NSFDC/PM-AJAY).
 - Working prototype with 90% of required functionality.
 - Fits hackathon resources: one web app, one API key, no new hardware.
- **Potential challenges and risks**
 - NCS, SIDH, PM-AJAY have no public developer APIs.
 - Live call needs connectivity and a Gemini key.
 - Catalog must stay current; must not collect Aadhaar / caste-cert / bank data.
- **Strategies for overcoming these challenges**
 - Seeded cited catalog now; MoSJE/SIDH DPI APIs when available.
 - Browser/kiosk for SIH; same script on IVR with a telecom partner.
 - Privacy rules in the system prompt.
 - Warn PM-DAKSH vs PM-AJAY double enrolment.

IMPACT AND BENEFITS

- **Potential impact on the target audience**
 - SC beneficiaries under PM-AJAY GIA with low literacy or dialect speech.
 - District GIA / SC corporation staff who lack ground counselling capacity.
 - Better matching is the missing step behind GIA's ~70% post-training placement rule.
- **Benefits of the solution (social, economic, environmental, etc.)**
 - **Social:** counselling in their language; traditional trades treated as real skills.
 - **Economic:** fewer wasted training seats; local wage jobs; NSFDC credit for self-employment.
 - **System:** each call becomes structured data for GIA perspective plans.

- **Details / Links of the reference and research work**

PM-AJAY GIA domains: pmajay.dosje.gov.in/illustrative-list

PM-AJAY portal: pmajay.dosje.gov.in

NSQF / NQR: nqr.gov.in

Kaushalverse: kaushalverse.ncvet.gov.in

SIDH: skillindiadigital.gov.in

PM-DAKSH: pmdaksh.dosje.gov.in

NCS jobs: ncs.gov.in

NSFDC: nsfdc.nic.in

e-Shram: eshram.gov.in

Gemini Live API: ai.google.dev/gemini-api/docs/live

PS 26097, MoSJE — Department of Social Justice and Empowerment.